ICCST Award in the Name of Wolfgang Gunther
Halle (Saale), Germany, August 2025

Wolfgang Günther loved chemistry. He fell in love with chemistry at 11 years old, when his father handed him his college level textbook Anorganische Chemie (Inorganic Chemistry) by K. A. Hoffmann (Vieweg, 1929). He read it cover-to-cover. To his mother's consternation, he began experimenting. In one infamous adventure, he nearly blew up her dining room trying to make rubies. He apparently had the temperature right, but had an issue with getting the right pressure.

I'm delighted to be here to join with you in honoring the memory of my father, Wolfgang Gunther (*editorial insert: the Umlaut ü was lost in transition to the USA*). My name is Bernard Gunther, and the ICCST conference has held an important place in my own life. My first direct connection with the conference was tagging along with my dad in Metz, France in 1979. Most recently, my wife and I were delighted to attend the 2019 ICCST conference with Wolfgang in Sardinia. It is a great honor to be with you today as we officially establish the award in my father's name.

Wolfgang was born in Gifhorn, Germany, in a country that was heading into the second world war, although no one knew that at the time. He was an only child, whose father spent much of his adolescence either as a soldier or as a prisoner of war.

Under intense competition for university admissions, he was accepted into the Foods Chemistry program at the Technische Universität Braunschweig, Germany. Three years into these studies, he was awarded a prestigious Foreign Office fellowship for study abroad and spent that at the University of Leeds, England. He spoke only the limited English he learned in school. The University found him a boarding house, where the lady of the house, Mrs. Barr, forbade him from speaking German, and made him speak "proper English".

He and his wife, Helge, met in England while both were pursuing their Ph.D. degrees. They traveled to Germany in 1958 to be married before they moved to the United States, traveling by boat. Wolfgang arrived in the US first, arriving on

the Hanseatic with $29 in his pocket. Two weeks later, Helge arrived, after a lucky win at Bingo allowed her to pay for offloading her luggage!

They both took up postdoc positions in the Pharmacology Department of Yale University Medical School in New Haven, Connecticut. Previously unfamiliar with the exact size of the US, Helge had originally considered a position at Montreal and Wolfgang at Yale, figuring that they could probably just live in Albany NY and commute. They had a one page map of Germany and a 1 page map of the US – so these cities look like they are close together. They are actually over 360 miles *(580 km)* apart.

In 1967, he joined Xerox in Webster, NY, in the US, leading the Organic Chemistry Research Area. In 1980, he moved to Eastman Kodak, leading their Dye Research Laboratory. He then got involved in Eastman Pharmaceuticals, and after several mergers and spin outs, retired from Sterling Winthrop in 1995. This didn't end his research. In fact, when we cleared out his house, we had to bring in a close friend and chemist to safely dispose of the chemistry lab he built in his basement.

Over his career, Wolfgang published over 70 books and papers. He was issued 177 patents globally. Quite an amazing set of accomplishments.

Wolfgang was a global citizen. Over his lifetime I counted that he visited over 45 countries.

It was science fiction writer William Gibson who famously said that "the future is already here, it just isn't evenly distributed yet." My father was someone who always lived partially in the future. He had traveled to the Palo Alto Research Laboratory in the 70's and came home with stories of computer mice and graphical user interfaces, which he then started using in his office in Webster NY. He would sketch out new molecular structures he wanted to explore while watching television at night. He used an Apple II computer and a Spinwriter printer to get around onerous corporate requirements many years before it became the norm. He was, in short, someone who was always pushing the boundaries of what was possible.

Wolfgang was a scientist, through and through. You never had to wonder if he would share his opinions, particularly if he thought there was a better way to do something. He was always willing to challenge your assumptions. Wolfgang wasn't trying to insist he was right, but rather that we should seek out the right answer.

Both my sister and I loved that we could chat about new ideas, science fiction concepts, chemistry, biology, and business. Whatever we were interested in, he would engage in the discussion. "What about X?" "But what if you did Y?" The discussions could get heated. My wife's early introduction to the family was after a dinner discussion, people scattering to grab a reference book to prove their case.

He lived in the future in many ways. Before there was such a thing as LinkedIn or the mass use of email lists, Wolfgang created the Sterlink network to keep the 300 people who left Sterling Chemicals connected. He created, printed, and mailed out directories with hundreds of names. Before there was Google, he regularly used my proprietary search engines. Before digital communications and email became commonplace, he had created his own early equivalents to use with his networks. And of course, he co-authored the definitive textbook on Selenium and Tellurium

His passion for the ICCST conference was unbounded. He co-founded the first conference of this series, which was held in New York in 1971. It was organized by physically mailing letters and waiting for weeks for people to respond. I get the impression that Wolfgang and his fellow organizers weren't certain if there would be broad interest among researchers in getting together and sharing ideas. He was thrilled to see how enduring the interest in the conference was. My mother's last international trip was to Gifu, Japan, in 2016 in connection with the conference there. It is profoundly moving to see how it has been taken up by new generations of scientists.

With my father passing, I don't think there is anyone left who has attended all sessions of this conference. He will remain with you, however, in the form of a funded grant and these awards that will give future scientists the kind of opportunities he was so grateful for.

What lessons can we draw from Wolfgang's life? These are lessons I hope our award recipients will endeavor to do and help others to do in the future.

- Be curious
- Seek out new inventions and creations
- Challenge yourself and be willing to be wrong
- Encourage others to do better
- Connect with others
- Enjoy life to the fullest and help others enjoy it

Our family is grateful to have been part of the ICCST community and look forward to seeing it thrive for many years to come.

Speech delivered by
Bernard Gunther *(for the Gunther family)*
on Friday, 29th August 2025 at ICCST-16,
held at the German National Academy of Sciences Leopoldina,
in Halle (Saale), Germany

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